s-family: internal patterns, pre-registered observable matching, structural links

Prepared 20 Oct 2026 (CDT). Local only: nothing was posted, the repo was not edited and nobody was contacted. All computations use PARI (cypari2), numpy and scipy.

Files:

- `partA_table.csv` (s = 5..500), `partA_analysis.out`, `partA_correlations.csv`.
- `PREREG.md`, with sha256 and timestamp in `prereg.sha256`. It was written before `partB_match.py` was run.
- `partB_results.json`, `partB_matches.csv`, `partB_posthoc.json`.

Part A is labelled Category A where proved, Category B otherwise. Part C is Category B, speculative.

Part A: internal patterns (s = 5..500)

The table was built with the manuscript's definitions (s01_intro.tex:131-134). The cusp dimension uses d_cusp = dim V - d_Eis, where:

- dim V is the orbit bound s(s-1) for odd s and (s/2)(s/2+1) for even s, reduced by m(m-1) when s = $2m^2$ (s12_dims.tex:8-15);
- d_Eis is the manuscript formula (s12_dims.tex:16-19);
- the manuscript value dim V = 2406 is used at s = 98.

Check: these formulas reproduce all 48 tabulated (dim V, d_Eis, d_cusp) rows exactly (s = 5..51 and 72). For s > 51 they are extrapolations, so Category B. The manuscript proves exactness only for s $\leq$ 51 and certifies d_Eis only for the listed local factors.

Proved (Category A)

- **When ε_s is not the fundamental unit.** Write ε_s = η^k with η the fundamental unit and t its trace. Then:
  - for N(η) = +1, ε_s = η^k exactly when s - 1 = T_k(t/2) (T_k the Chebyshev polynomial);
  - for N(η) = -1, the analogous condition holds with the norm -1 trace.

This is standard: if $\eta = (t+u\sqrt{d})/2$, then $\eta^k + \eta^{-k} = 2T_k(t/2)$. The 34 cases with k > 1 for s $\leq$ 500 are listed in `partA_analysis.out`, and every one satisfies the condition with the computed integer trace. The cases fall into these families:

  - k = 2, N = +1: s = $2m^2$, for example 8, 32, 50, 72, 128 (the manuscript's mechanism, s12_dims.tex:14-15). At s = 18, 98 and 162 the power is 4 or 12, not 2.
  - k = 2, N = -1: s = $2m^2$ + 2, i.e. 20, 34, 52, 74, ...
  - k = 3: s - 1 = (t^3 - 3t)/2, i.e. s = 27, 56, 245, 352, 486.
  - k = 5: s = 363 (t = 4).
  - k = 6 or 12: s = 10, 100, 162.

In total ε_s is fundamental for 462 of 496 values of s.

- **Genus theory.** 2^ω(d_K)-1 divides h^+(K) for every s, verified 100%. It follows that h^+ is divisible by 4 on the twin-prime subfamily (d_K = 4(s-2)s, ω = 3), as observed for all 24 twin-prime s up to 500.
- **Closed form for odd s,** given the manuscript's Theorem dims (iii) and dim V = s(s-1): d_cusp = s(s-1) - 3$\prod$_p^e$\|$n_s(p^e+1)/2.

Empirical (Category B)

- **Zero cusp part:** d_cusp = 0 only at s = 6, 8, 10, both in the tabulated range and in the extrapolation to 500.
- **Parity:** d_cusp is even for every s $\leq$ 500. This is not proved here.
- **Growth:** d_cusp/dim V tends upward. log(ratio) $\approx$ 0.24 log s + const for odd s (r = 0.55) and 0.33 log s + const for even s (r = 0.64). The median ratio for s $>$ 200 is 0.59 (odd) and 0.61 (even). This is a trend, not a closed-form law.
- **Cohen–Lenstra:** among the 479 distinct fields, 53.9% have trivial odd part of h. The Cohen–Lenstra heuristic for real quadratic fields gives about 75.4% (95% half-width 3.9%), so these fields depart from it. That is expected rather than anomalous: d = m^2 - 1 (m = s - 1) is Richaud–Degert type, with an unusually small regulator (ε_s $\approx 2\sqrt{d}$), so class numbers run larger (Yokoi; Mollin, *Quadratics*, 1996). The heuristic assumes generic regulators.
- **Correlations:** 36 Spearman tests (raw, and with the log-s trend removed), with 2000-sample permutation p-values and BH correction; see `partA_correlations.csv`. Every BH-significant correlation follows from known relations:
  - h^+ against the ray-class number (ρ = 0.91): the ray-class exact sequence.
  - h^+ against the conductor f (ρ = -0.85 detrended): the class number formula h·R ~ $\sqrt{d_K}$, with d_K = 4n_s/f^2.
  - regulator against k (ρ = -0.97 detrended): R = log ε_s / k.
  - correlations with s itself: Brauer–Siegel growth.

No correlation was found that is unexplained.

Part B: pre-registered observable matching

Setup (PREREG.md):

- Quantities: Q1 = 2 log ε_s, Q2 = I_s, Q3 = h^+/h, Q4 = d_cusp/dim V (tabulated s), Q5 = $\sqrt{n_s}$. That is 2032 values.
- Operations: x, 1/x, 2x, 3x, x/2 and x/3.
- 16 cited targets.
- **N = 195 072 trials.**
- Criterion: |f(x) - y| $\leq$ max(σ, 10^{-3}|y|).

Deviation from the pre-registration, disclosed: T15's σ was recomputed from the cited inputs as 54.6 ($\times 10^{-8}$), not the 58.4 typed in PREREG. That is an arithmetic slip in the pre-registration and does not change any conclusion.

Pre-registered result

- **Observed M = 10 633 matches.**
- Null means (10^4 trials each):
  - N1 log-uniform: 11 903 $\pm$ 45, p = 1.0.
  - N2 random-field units: 8822 $\pm$ 44, p $\approx$ 1e-4.
  - N3 random integers: 10 505 $\pm$ 23, p $\approx$ 1e-4.
- **The global count is dominated by two pre-registered targets that turned out to be degenerate:**
  - T15 (g-2 deviation 33 $\pm$ 55): any value in [-22, 88] matches, giving 10 067 of the matches.

  - T16 (m_W CDF/PDG = 1.0008, with tolerance 10^{-3}): any value equal to 1 matches, so the 496 values of Q3 = 1 all match.

- The low p-values under N2/N3 come from differences between those nulls and the s-family value distribution inside the wide T15 window. They do not reflect precise matches. The p-values disagree across nulls (1.0 against 1e-4), which shows the global statistic is null-model-dependent and uninformative.

Post-hoc (labelled; `partB_posthoc.py`)

Excluding T15 and T16:

	Count
Observed	67
N1	172.8 ± 13.1 (p = 1.0)
N2	90.1 ± 9.5 (p = 0.99)
N3	145.3 ± 11.6 (p = 1.0)
N4 (jitter: each true value × e^N(0,0.02), closest-matched distribution)	69.2 ± 6.1 (p = 0.67)

The s-family produces no more matches than distribution-matched random numbers.
Per target:

- No target is significant under N4.
- T1 (α^{-1}) had p = 0.04 under N1 only. With 16 × 4 tests that is not significant, and under N2, N3 and N4 the p-values are 0.22–0.48.
- **None of the 67 matches is within 1σ for any precision target (T1–T5).** They pass only through the 10^{-3} relative tolerance.

Examples:

- $\sqrt{}$n_s at s = 138 is 136.996, against $\alpha^{-1} = 137.036$: a relative difference of 2.9e-4, i.e. 1.9×10^6 σ. This is a trivial near-integer, since $\sqrt{}$(s(s-2)) ≈ s - 1.
- 1/I_s at s = 470 is 137.059 (1.7e-4, about 10^6 σ).
- 1/I_s at s = 37 is 16.830, against m_τ/m_μ = 16.817 (0.08%, 12σ).
- (2/2)·d_cusp/dim V at s = 41, giving 0.2311, against $\sin^2\theta$_W = 0.23129 (0.08%, 5σ).

Verdict for Part B: chance.
The value s = 7 itself matches nothing: 2 log ε_7 = 4.9558, I_7 = 0.20942, $\sqrt{}$35 = 5.916, d_cusp/dim V = 1/7.

Part C: structural correspondences (Category B, speculative)

- **Weil representation as a quantum cat map (most promising).**
  - γ_s = [[-1,-s],[2,2s-1]] is a hyperbolic $SL_2(Z)$ element with entropy log ε_s, an Anosov cat map.

  - Its quantization is given by the Weil (metaplectic) representation: Hannay & Berry, Physica D 1 (1980) 267; Kurlberg & Rudnick, Duke Math J 103 (2000) 47 (Hecke eigenbases).
  - The Bridge C label space carries exactly a Weil representation of the discriminant form (s03_main.tex:77-87; s12_dims.tex proof notes).
  - **Next computation:** the spectrum of ρ_s(γ_s) on W_P for s = 5..60. Compare the eigenphase statistics and the period of ρ_s(γ_s) mod the label level with Kurlberg–Rudnick predictions. Test whether the cusp sector V_cusp coincides with an anti-invariant or Hecke-eigen subspace.

- **Indefinite theta series and wall-crossing / two-centred black-hole bound states.**
  - Signature-(1,1) Zwegers theta functions count two-centre D4-D2-D0 bound states: Manschot, arXiv:0906.1767 and arXiv:1009.1775; Alexandrov–Banerjee–Manschot–Pioline, arXiv:1606.05495 (generalized error functions).
  - The mock modularity of BH microstate generating functions: Dabholkar–Murthy–Zagier, arXiv:1208.4074.
  - **Next computation:** identify whether a lattice $\langle 2\rangle \oplus \langle$-2(s-2)s$\rangle$-type form, with the cone walls c_1 = (-1,1) and c_2 = (-4,1), arises as the charge lattice of a two-centre configuration (some CY_3 intersection form, with D = s(s-2)). If it does, compare H_s with the corresponding index jump across the wall.

- **Quantum modularity and 3-manifold invariants.**
  - The radial and rational values ψ(h/m) of the Bridge C period function (s07_lz.tex:13) are cyclotomic. Quantum modular forms arising from false and indefinite theta series give WRT and $\hat{Z}$ invariants: Zagier, "Quantum modular forms" (Clay Math Proc. 12, 2010); Lawrence–Zagier, Asian J Math 3 (1999) 93; Gukov–Pei–Putrov–Vafa, arXiv:1701.06567; Cheng–Chun–Ferrari–Gukov–Harrison, arXiv:1809.10148.
  - **Next computation:** test whether ψ(h/m) at s = 7 satisfies a quantum-modular cocycle with real-analytic error. Search the plumbed or Seifert manifolds whose $\hat{Z}$ uses an indefinite lattice of determinant -n_s.

Also noted but weaker:

- K3 lattices and Mathieu moonshine (Eguchi–Ooguri–Tachikawa, arXiv:1004.0956). The manuscript's K3 section addresses only lattice embedding; no elliptic genus appears.

None of the three is an observable prediction. They are mathematical correspondences that could become physics only through such a construction.